



STEM SMART Single Maths 28 - Binomial Theorem - Rational n >

Binomial Expansion - Rational n

A-level Maths Topic Summaries - Series

Subject & topics: Maths | Algebra | Series **Stage & difficulty:** A Level P3

Fill in the blanks below to complete the notes on binomial expansion for rational powers of n .

Brackets of the form $(1 + x)^n$, where n is any number, can be expanded using the expression

$$(1 + x)^n = 1 + \text{} x + \frac{n(n-1)}{2!} \text{} + \frac{n(n-1)(n-2)}{3!} \text{} + \dots$$

The right hand side is a sum of positive integer powers of x , plus a constant. Sums of this form are called .

The right hand side has an infinite number of terms. The right hand side converges to the left hand side as long as .

If $|x|$ is small, we can approximate $(1 + x)^n$ using the terms of the series. The terms we use, the better our approximation will be.

Items:

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Question deck:

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Binomial: All Rational n 2ii

Subject & topics: Maths Stage & difficulty: A Level P2

Part A

Expansion

Expand $(1 + 4x)^{\frac{1}{2}}$ in ascending powers of x , up to and including the term in x^2 , simplifying the coefficients.

The following symbols may be useful: x

Part B

Values of x

State the set of values of x for which the expansion is valid. You may use the symbols $<$ $>$ $=$ x and $|x|$ which can be entered as `abs(x)`.

The following symbols may be useful: $<$, $>$, `abs()`, x

Part C

Value of k

In the expansion of

$$(1 + kx)(1 + 4x)^{\frac{1}{2}},$$

the coefficient of x is 7. Find the value of the constant k and hence the coefficient of x^2 .

$k =$

coefficient of $x^2 =$

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Binomial: All Rational n 4ii

Subject & topics: Maths **Stage & difficulty:** A Level P2

Part A

Expansion

Expand $(2 + x)^{-2}$ in ascending powers of x up to and including the term in x^3 .

The following symbols may be useful: x

Part B

Set of Values

State the set of values of x for which the expression is valid. You may use the symbols $<$ $>$ $=$ x and $|x|$ which can be entered as `abs(x)`.

The following symbols may be useful: $<$, $>$, `abs()`, x

Part C

Coefficient

Hence find the coefficient of x^3 in the expansion of $\frac{1+x^2}{(2+x)^2}$.

The following symbols may be useful: x

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Binomial: All Rational n 1i

Subject & topics: Maths

Stage & difficulty: A Level P2

Part A

Partial Fractions

Given that $\frac{3x + 4}{(1 + x)(2 + x)^2} \equiv \frac{A}{1 + x} + \frac{B}{2 + x} + \frac{C}{(2 + x)^2}$, find A , B , and C .

If a value is not a whole number, enter the value as a decimal.

$A =$

$B =$

$C =$

Part B

Expand

Hence or otherwise expand $\frac{3x + 4}{(1 + x)(2 + x)^2}$ in ascending powers of x , up to and including the term in x^2 .

The following symbols may be useful: x

Part C

Values of x

State the set of values of x for which the expansion in the above part is valid.

Construct your answer from the items below.

Items:

x

$<$

$\leq$

$>$

$\geq$

$< x <$

$\leq x \leq$

$> x \text{ or } x >$

$\geq x \text{ or } x \geq$

-4

-2

$-\frac{3}{2}$

-1

$-\frac{1}{2}$

0

$\frac{1}{2}$

1

$\frac{3}{2}$

2

4

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Binomial: All Rational n 3i

Subject & topics: Maths **Stage & difficulty:** A Level P2

Find the coefficient of x^2 in the expansion in ascending powers of x of

$$\sqrt{\frac{1+ax}{4-x}}$$

giving your answer in terms of a .

The following symbols may be useful: a

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Binomial Expansion 5

Pre-Uni Maths for Sciences C2.7

Subject & topics: Maths | Algebra | Series **Stage & difficulty:** A Level P3

Rewrite each of the following expressions in the form $a(1 + b)^n$, where a is an integer and b is a fraction such that $|b| < 1$. Hence, using the appropriate binomial expansion, find the value of each of them correct to 4 dp.

Part A

$$\sqrt{36.1}$$

Rewrite $\sqrt{36.1}$ in the form $a(1 + b)^n$, where a is an integer, b is a fraction such that $|b| < 1$, and $n = \frac{1}{2}$.

$$\sqrt{36.1} = \boxed{} (1 + \boxed{})^{\frac{1}{2}}$$

Hence, using the appropriate binomial expansion, find the value of $\sqrt{36.1}$ correct to 4 dp.

$$\sqrt{36.1} = \boxed{}$$

Part B

$$\sqrt[3]{1.09}$$

Rewrite $\sqrt[3]{1.09}$ in the form $a(1 + b)^n$, where a is an integer, b is a fraction such that $|b| < 1$ and $n = \frac{1}{3}$.

$$\sqrt[3]{1.09} = \boxed{} (1 + \boxed{})^{\frac{1}{3}}$$

Hence, using the appropriate binomial expansion, find the value of $\sqrt[3]{1.09}$ correct to 4 dp.

$$\sqrt[3]{1.09} = \boxed{}$$

Part C

$$\frac{1}{\sqrt{1.04}}$$

Rewrite $\frac{1}{\sqrt{1.04}}$ in the form $a(1+b)^n$, where a is an integer, b is a fraction such that $|b| < 1$ and $n = -\frac{1}{2}$.

$$\frac{1}{\sqrt{1.04}} = \boxed{} (1 + \boxed{})^{-\frac{1}{2}}$$

Hence, using the appropriate binomial expansion, find the value of $\frac{1}{\sqrt{1.04}}$ correct to 4 dp.

$$\frac{1}{\sqrt{1.04}} = \boxed{}$$

Part D

$$\sqrt[3]{125.4}$$

Rewrite $\sqrt[3]{125.4}$ in the form $a(1+b)^n$, where a is an integer, b is a fraction such that $|b| < 1$ and $n = \frac{1}{3}$.

$$\sqrt[3]{125.4} = \boxed{} (1 + \boxed{})^{\frac{1}{3}}$$

Hence, using the appropriate binomial expansion, find the value of $\sqrt[3]{125.4}$ correct to 4 dp.

$$\sqrt[3]{125.4} = \boxed{}$$

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Maclaurin Series - Potential due to Dipole

Pre-Uni Maths for Sciences C3.9

Subject & topics: Maths | Algebra | Series

Stage & difficulty: Further A C3, University C1

Part A

Potential on the axis

An electric dipole consists of two charges $+q$ and $-q$ separated in the z direction by a very small distance a . The electric potential $V(z)$ a distance z away from the centre of the dipole in a direction along the line joining the two charges is given by

$$V(z) = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{z - \frac{a}{2}} - \frac{1}{z + \frac{a}{2}} \right).$$

Find an approximate expression for $V(z)$ (assume that $z \gg a$ and obtain the first non-zero term in the Maclaurin (or binomial) expansion of $V(z)$).

The following symbols may be useful: a , ϵ_0 , π , q , z

Part B

Potential in any direction

An electric dipole consists of two charges $+q$ and $-q$ separated by a distance a . The electric potential V a distance r from the centre of the dipole in a direction making an angle θ to the line joining the two charges is given approximately by

$$V \approx \frac{q}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{r^2 - ar \cos \theta}} - \frac{1}{\sqrt{r^2 + ar \cos \theta}} \right).$$

Assuming that $r \gg a$ show that $V \approx \frac{A \cos \theta}{r^2}$ and find an expression for A .

The following symbols may be useful: a , ϵ_0 , π , q , r

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